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ON PATH DECOMPOSITIONS OF GRAPHS

by

Lirong Yan

A Dissertation Presented in Partial Fulfillment
of the Requirements for the Degree
Doctor of Philosophy

ARIZONA STATE UNIVERSITY

May 1998

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March 1998

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ABSTRACT

Let $G = (V, E)$ be a finite,undirected graph without loops or multiple edges. A path decomposition of G is a partition of the edges of G into paths. Such a decomposition is said to be minimal if its cardinality is no greater than the cardinality of any other such partition. The path number $\varphi(G)$ of G is the cardinality of a minimal path decomposition of G.

Gallai conjectured that if G is a connected graph with n vertices, then $\varphi(G) \leq \lfloor \frac{n+1}{2} \rfloor$. The author proves that if G is a graph with n vertices (not necessarily connected), then its path number $\varphi(G)$ is less than or equal to $\frac{1}{2}\mu + \lfloor \frac{2}{3} g \rfloor$, where μ is the number of vertices of odd degree and g is the number of nonisolated vertices of even degree.

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DEDICATION

To my husband, my parents, my son and my daughter who gave me their love and support.

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I am particularly grateful for the guidance and patience of my advisor, Professor Genghua Fan. Without his meaningful criticisms, suggestions and insights, this dissertation would not have been possible. I also would like to thank my committee members, Professor Hal Kierstead, Professor Glenn Hurlbert, Professor Hélène Barcelo, and Professor Ed Ihrig for their help with this dissertation.

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Chapter 1

INTRODUCTION

1.1 Introduction

Let $G = (V, E)$ be a finite, undirected graph without loops or multiple edges. The *vertex-set* and the *edge-set* of G are denoted by $V(G)$ and $E(G)$, respectively. For $v \in V(G)$, $N_G(v)$ is the set, and $d_G(v)$ is the number, of neighbors of v in G. Moreover, $N_e(v)$ ($N_o(v)$) is the set of *even* (*odd*) neighbors of v in G. For $u, v \in V(G), d_G(u, v)$ denotes the distance between u and v in G.

A *path* (*path-circuit*) *decomposition* of a graph G is a partition of the edges of G into paths (paths and circuits). A path decomposition is said to be *minimal* if its cardinality is no greater than the cardinality of any other such partition. The *path number* of G, denoted by $\wp(G)$, is the cardinality of a minimal path decomposition of G.

In this dissertation our goal is to study path decompositions of graphs. We shall prove that if a graph G has μ vertices of odd degree and g nonisolated vertices of even degree, then its path number is less than or equal to $\frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor$, which is best possible.

Now we briefly present the background and summarize the results concerning path decompositions of graphs.

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Erdös asked what the minimal number of disjoint paths was with which we could partition any connected graph of n vertices. Gallai conjectured that this number was $\lfloor \frac{n+1}{2} \rfloor$. Therefore, we have the following:

Conjecture 1.1.1 *(Gallai's Conjecture) If G is a connected graph with n vertices, then*

$$\varphi(G) \leq \lfloor \frac{n+1}{2} \rfloor.$$

Suppose G has k nontrivial components $G_1, G_2, \ldots, G_k$ each with an odd number of vertices. If G_i has n_i vertices for $i = 1, 2, \ldots, k$, then $n_i \geq 3$ so

$$n = \Sigma_{i=1}^{k} n_i \geq 3k.$$

Thus Gallai's conjecture implies that

$$\begin{aligned} \varphi(G) &\leq \Sigma_{i=1}^{k} \lfloor \tfrac{n_i+1}{2} \rfloor \\ &= \Sigma_{i=1}^{k} \tfrac{n_i}{2} + \tfrac{k}{2} \\ &= \tfrac{1}{2}n + \tfrac{1}{2}k \\ &\leq \tfrac{2}{3}n. \end{aligned}$$

Therefore, $\varphi(G) \leq \lfloor \frac{2}{3}n \rfloor$.

Now, consider a graph G with n vertices each of whose component is a triangle. Then

$$\varphi(G) = \lfloor \frac{2}{3}n \rfloor (= \frac{2}{3}n).$$

So if Gallai's conjecture is true, then the best bound possible for the path number of a graph (not necessarily connected) G with n vertices is $\lfloor \frac{2}{3}n \rfloor$. This dissertation is motivated by this observation.

Although Conjecture 1.1.1 appears to be very difficult and has survived for decades, there have been some interesting and strong partial results. The following classical result was proved by Lovász [10] in 1968.

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Theorem 1.1.1 *(Lovász Theorem) If G is a graph (not necessarily connected) with n vertices, then there exists a partition $\wp$ of the edges of G into paths and circuits with*

$$|\wp| \leq \lfloor \frac{1}{2} n \rfloor.$$

Let a graph G have μ odd vertices and g even vertices, $\mu + g = n$. If G has at most one even vertex, that is, $g = 0, 1$, Lovász theorem says that we may partition the graph G into $\leq \lfloor \frac{1}{2} n \rfloor = \frac{1}{2}\mu + \lfloor \frac{1}{2} g \rfloor = \frac{1}{2}\mu$ disjoint paths and circuits. Since every odd vertex must be the endpoint of at least one path, so there is no circuit. Therefore we have the following Corollary:

Corollary 1.1.1 *If G is a graph with n vertices of odd degree only, then*

$$\wp(G) = \frac{1}{2} n.$$

If $g \geq 1$, let us add $g - 1$ new vertices to G, each joined to a distinct even vertex of G. If we decompose the obtained graph G' into $\lfloor \frac{n+g-1}{2} \rfloor = \frac{1}{2}\mu + g - 1$ disjoint paths and circuits, then this decomposition uses only paths (since G' has only one even vertex). Thus we have

Corollary 1.1.2 *If G is a graph containing exactly μ odd vertices and $g \geq 1$ nonisolated even vertices, then*

$$\wp(G) \leq \frac{1}{2}\mu + g - 1.$$

As an extension of Corollary 1.1.1, L.Pyber [11] proved that

Theorem 1.1.2 *(Pyber Theorem) Suppose that each cycle of G contains a vertex of odd degree. Then G can be partitioned into $\leq \lfloor \frac{n}{2} \rfloor$ edge-disjoint paths.*

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Consider a graph G obtained from the complete graph K_{2m+1} by the deletion of $m-1$ independent edges. Then G has exactly $(n-1)\lfloor\frac{n}{2}\rfloor+1$ edges ($n=2m+1$) and therefore we need $\lfloor\frac{n}{2}\rfloor+1$ paths. On the other hand, the only cycle of G with all degrees even is a triangle. Therefore Theorem 1.1.2 is best possible.

Corollary 1.1.2 was strengthened by Donald [6] who proved that

Theorem 1.1.3 *(Donald Theorem) Let G be a graph with μ vertices of odd degree and g nonisolated vertices of even degree. Then*

$$\varphi(G) \leq \frac{1}{2}\mu + \lfloor\frac{3}{4}g\rfloor.$$

We shall improve this upper bound and establish the following theorem, which is the main result of this dissertation.

Main Theorem Let G be a graph with μ vertices of odd degree and g nonisolated vertices of even degree. Then

$$\varphi(G) \leq \frac{1}{2}\mu + \lfloor\frac{2}{3}g\rfloor.$$

1.2 Related Topics

It seems very difficult to prove Gallai's conjecture. Chung [5] has suggested the investigation of the weaker version in which the paths are not necessarily edge-disjoint. Let φ be a collection of paths of G. If each edge $e \in E(G)$ is contained in at least one path of φ, we say that G is covered by φ.

Chung [5] proposed the following conjecture:

Conjecture 1.2.1 *(Chung's Conjecture) A connected graph G with n vertices can be covered by $\lceil\frac{n}{2}\rceil$ simple paths.*

In an attempt to prove Chung's conjecture, Pyber [11] obtained the following Theorems.

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Theorem 1.2.1 *Every connected graph G on n vertices can be covered by $\frac{n}{2}+O(n^{\frac{3}{4}})$ paths.*

Theorem 1.2.2 *Every connected graph on n vertices with e edges can be covered by $\frac{n}{2}+4(\frac{e}{n})$ paths.*

Note that the paths in Theorem 1.2.1 and Theorem 1.2.2 are not necessarily edge-disjoint. And Chung's conjecture is still open.

A *weighted graph* (G,ω) is a graph G with a weight function $\omega : E(G) \to \{0,1,2\}$. If v is a vertex of (G,ω), define $\omega(v) = \Sigma_{u\in N(v)}\omega(uv)$ and call v an *odd (even)* vertex if $\omega(v)$ is odd (even). A *path cover* of (G,ω) is a collection of paths in G such that each edge e is contained in exactly $\omega(e)$ paths of the collection.

Fan [8] studied path covers of weighted graphs and proved that

Theorem 1.2.3 *If every vertex of (G,ω) is incident with at most one edge of weight 2, then (G,ω) has a path cover $\wp$ such that each odd vertex occurs exactly once, and each even vertex exactly twice, as an end of a path of $\wp$.*

Theorem 1.2.4 *If every vertex of (G,ω) is even, then (G,ω) has a path cover such that each vertex occurs exactly twice as an end of a path of $\wp$.*

Note that Corollary 1.1.1 is also a corollary of Theorem 1.2.3 by taking $\omega(e) = 1$ for all $e \in E(G)$.

For a graph G, let $\omega(e) = 2$ for all $e \in E(G)$. Then by Theorem 1.2.4, G has a collection $\wp$ of paths such that each edge is contained in exactly two paths of $\wp$ and each vertex occurs exactly twice as an end of a path of $\wp$. This was known as the Perfect Path Double Cover Conjecture, proposed by Bondy [1] and settled by Li [9].

Let $G = (V,E)$ be a multigraph. For any two vertices x and y of G, $\rho_G(x,y)$ denotes the number of (multiple) edges joining x and y. For each vertex x of G,

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$\mu_G(x) = max\{\rho_G(x,y) : y \in N_G(x)\}$ specifies the *multiplicity* of x, and $d_G(x)$ denotes the *degree* (the total number of edges incident with x) of x. For any integer k, let $\lceil k \rceil_e$ be the least even integer $\geq k$ and $\lceil k \rceil_o$ the least odd integer $\geq k$. Define a function of a vertex $x \in V$ as follows:

$$\phi_G(x) = \begin{cases} \lceil \mu_G(x) \rceil_e & , \quad \text{if } x \text{ is an even vertex} \\ \lceil \mu_G(x) \rceil_o & , \quad \text{otherwise} \end{cases}$$

A path decomposition φ of G is a *faithful path decomposition* (FPD) if every vertex x of G is an end of exactly $\phi_G(x)$ paths in φ.

Cai [4] obtained the following interesting result that can be used to deduce Corollary 1.1.1, Theorem 1.2.4, and some other results on path decompositions.

Theorem 1.2.5 *Every multigraph admits a faithful path decomposition.*

1.3 Organization

We now outline the next two chapters of this dissertation. The main purpose of Chapter 2 is to review and establish some fundamental techniques which are essential to the proof of the main theorem in Chapter 3.

In Section 2.1 we review Lovász construction [10]. In Section 2.2 we introduce the extention of Lovász construction.

Chapter 3 gives a complete proof of the main theorem. In Section 3.1 we present some interesting observations on Lovász construction. In Section 3.2 we prove some fundamental lemmas based on Lovász construction and the extention of Lovász construction. The proof of the main theorem is given in Section 3.3.

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Chapter 2

FUNDAMENTAL TECHNIQUES

The goal of this chapter is to introduce and develop some fundamental techniques which provide us some important lemmas for the main theorem in the next chapter.

2.1 Lovász Construction

Let G be a graph with at least one nonisolated even vertex and let x be a vertex adjacent to an even vertex. Denote by $a_1, a_2, \ldots, a_k$ the even vertices of G adjacent to x. Let G' be the graph formed from G by deleting the edges $xa_i, i = 1, 2, \ldots, k$. Then every vertex in $N_G(x)$ is odd in G'. Let Σ be a minimal path decomposition of the edges of G'.

For each $i = 1, 2, \ldots, k$, we define a sequence of vertices of $N_G(x)$: $a_{i,0}, a_{i,1}, \ldots, a_{i,r_i}$ by recursion as follows. Let $a_{i,0} = a_i$. Suppose $a_{i,\mu}$ is defined. Since $a_{i,\mu} \in N_G(x)$, it is of odd degree in G', so there is a path $U_{i,\mu}$ of Σ that has $a_{i,\mu}$ as an endpoint. If $U_{i,\mu}$ does not pass through x, end the sequence; If x lies on $U_{i,\mu}$, trace the path from $a_{i,\mu}$ to x and let $a_{i,\mu+1}$ be the last vertex on the path before it reaches x. We call each sequence

$$S_i = (a_{i,0}, a_{i,1}, \ldots, a_{i,r_i})$$

a Lovász sequence.

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Lovász [10] and Donald [6] make the following observations on the Lovász sequences:

Observation 2.1.1 If $a_{i,\mu} = a_{j,\nu}$, then $i = j$ and $\mu = \nu$.

For suppose $a_{i,\mu} = a_{j,\nu}$ and $\mu \leq \nu$. If $\mu = 0$, then $a_{i,\mu} = a_i$, so $\nu = 0$ and $i = j$. If $\mu \geq 1$, then $a_{i,\mu}$ lies on $U_{i,\mu-1}$, which has an endpoint at $a_{i,\mu-1}$ and contains the edge $a_{i,\mu}x$. Similarly, $a_{i,\mu} = a_{j,\nu}$ lies on $U_{j,\nu-1}$ which contains $a_{j,\nu}x = a_{i,\mu}x$. Since the paths of Σ are edge disjoint, $U_{i,\mu-1} = U_{j,\nu-1}$ and $a_{i,\mu-1} = a_{j,\nu-1}$. If we continue this tracing process, we arrive at $a_{i,0} = a_{j,\nu-\mu}$. By the same argument used for the case $\mu = 0$, we have $\nu - \mu = 0$. So $\mu = \nu$ and $i = j$.

This implies that Lovász sequences are finite.

Observation 2.1.2 The labeling of paths associated with the Lovász sequences is not unique: we can have $U_{i,\mu} = U_{j,\nu}$ with $(i,\mu) \neq (j,\nu)$.

Observation 2.1.3 $a_{i,\mu}$ is adjacent to x in G' iff $\mu \geq 1$. Furthermore, the edge $a_{i,\mu}x$ lies on $U_{i,\mu-1}$ iff $\mu \geq 1$.

We now define a function f that maps elements of Σ into paths and circuits that partition the edges of G. Let U be a path of Σ. If U is not a $U_{i,\mu}$, let $f(U) = U$. Otherwise we distinguish two sets of pairs of mutually exclusive cases.

The first pair is

(a) $U = U_{i,\mu}$ contains x, and

(a′) $U = U_{i,\mu}$ does not contain x.

The second pair is

(b) $U = U_{i,\mu} = U_{j,\nu}$ where $(i,\mu) \neq (j,\nu)$, and

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(b′) $U = U_{i,\mu}$ and the labeling is unique.

We combine these cases into four possible situations and define $f(U)$ for each.

Case(ab) (a) and (b) hold (see Fig. 2.1 case(ab)).

Let $f(U_{i,\mu} = U_{j,\nu})$ be $U_{i,\mu} = U_{j,\nu}$ with the edges $xa_{i,\mu+1}$ and $xa_{j,\nu+1}$ deleted and the edges $xa_{i,\mu}$ and $xa_{j,\nu}$ added.

Case(ab′) (a) and (b′) hold (see Fig. 2.1 case(ab′)).

Let $f(U_{i,\mu})$ be $U_{i,\mu}$ with the edge $xa_{i,\mu+1}$ deleted and the edge $xa_{i,\mu}$ added.

Case(a′b) (a′) and (b) hold (see Fig. 2.1 case(a′b)).

Let $f(U_{i,\mu} = U_{j,\nu})$ be the circuit obtained by adding $xa_{i,\mu}$ and $xa_{j,\nu}$ to $U_{i,\mu} = U_{j,\nu}$. Note that case(a′b) produces a circuit and this is the only case in which a circuit is formed.

Case(a′b′) (a′) and (b′) hold (see Fig. 2.1 case(a′b′)).

Let $f(U_{i,\mu})$ be the path obtained by adding $xa_{i,\mu}$ to $U_{i,\mu}$.

This completes the definition of f. We now show that the paths and circuits formed by f are a partition of the edges of G; i. e., if yz is an edge of G, then yz appears in exactly one $f(U)$. Clearly, if yz is not of the form $xa_{i,\mu}$, then yz appears in some U of Σ and also in, and only in, f(U). It is obvious from an examination of the cases that $xa_{i,\mu}$ appears in $f(U_{i,\mu})$, so it remains to prove only that if $xa_{k,\rho}$ is in $f(U_{i,\mu})$, then $U_{k,\rho} = U_{i,\mu}$.

Let $xa_{k,\rho}$ be an edge of $f(U_{i,\mu})$ and assume that $a_{k,\rho} \neq a_{i,\mu}$. $U_{i,\mu}$ falls under one of the four cases mentioned above.

If $U_{i,\mu}$ is of the form (ab) or (a′b), then $U_{i,\mu} = U_{j,\nu}$, where $(i,\mu) \neq (j,\nu)$. The only edges of $f(U_{i,\mu})$ incident to x are $xa_{i,\mu}$ and $xa_{j,\nu}$. Since $a_{k,\rho} \neq a_{i,\mu}$, by assumption, it follows that $a_{k,\rho} = a_{j,\nu}$, so $(k,\rho) = (j,\nu)$ by Observation 2.1.1.

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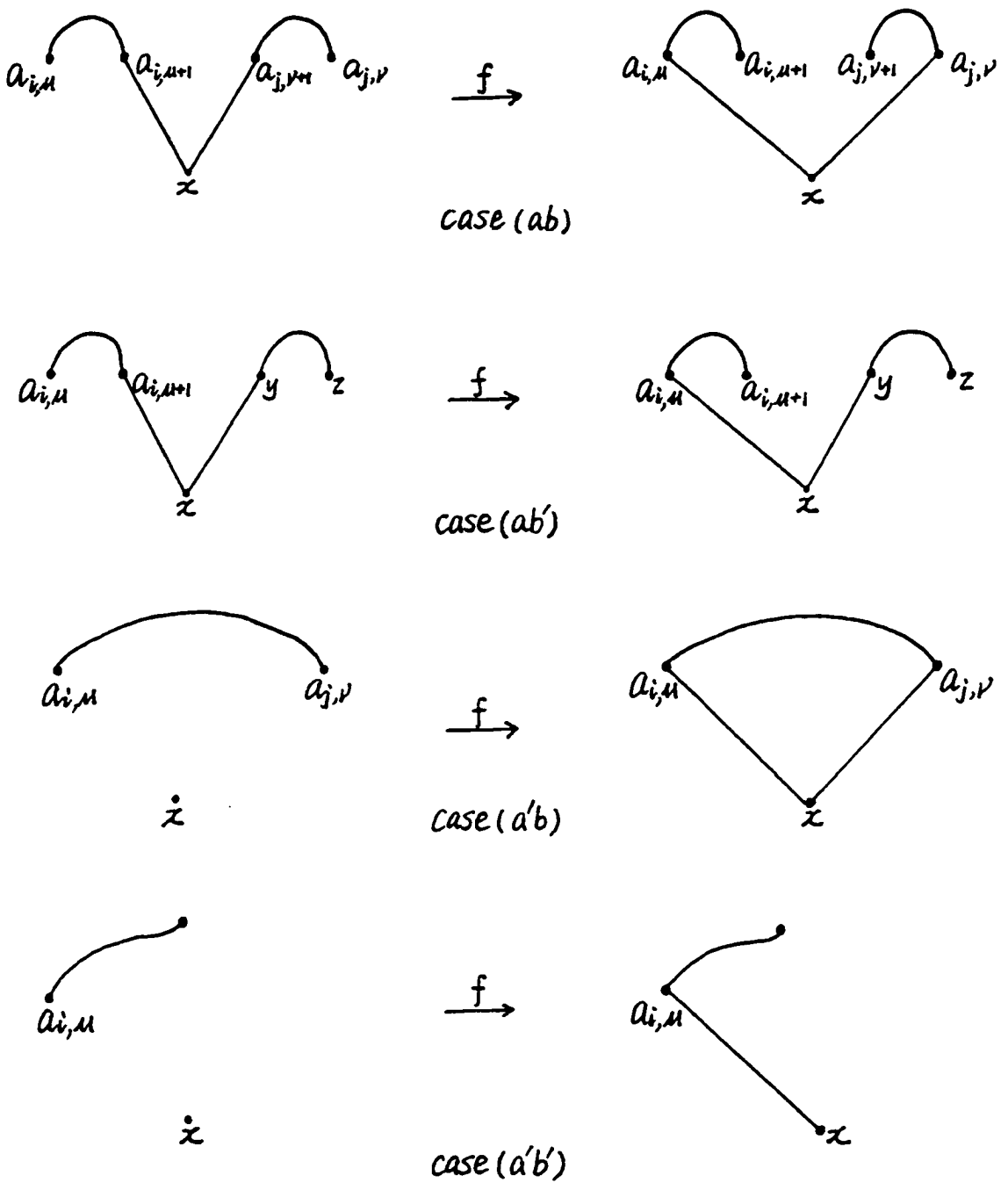

Figure 2.1: The definition of a function f in Lovász construction

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If $U_{i,\mu}$ is of the form (ab′), let $U_{i,\mu} = (a_{i,\mu}, \ldots, a_{i,\mu+1}, x, y, \ldots, z)$. (See Fig. 2.1 case(ab′).) If $xa_{k,\rho}$ is in $f(U_{i,\mu})$ and $a_{k,\rho} \neq a_{i,\mu}$, then $a_{k,\rho} = y$. Thus $z = a_{k,\rho-1}$ and $U_{i,\mu} = U_{k,\rho-1}$. But $U_{i,\mu}$ is of type (b′) so it is uniquely labeled. So $(i, \mu) = (k, \rho-1)$ and $a_{i,\mu} = a_{k,\rho-1} = z$. Thus $U_{i,\mu}$ must have been a circuit, which is not the case. So $xa_{k,\rho} \notin f(U_{i,\mu})$ unless $(k, \rho) = (i, \mu)$.

If $U_{i,\mu}$ is of the form (a′b′), the only edge of $f(U_{i,\mu})$ incident to x is $xa_{i,\mu}$.

Thus $f(\Sigma)$ is a partition of the edges of G into paths and circuits and $|f(\Sigma)| = |\Sigma|$.

We call the above construction the Lovász construction about vertex x.

Let G be a graph with at least one nonisolated even vertex and let x be a vertex adjacent to an even vertex. Denote by $a_1, a_2, \ldots, a_k$ the even vertices of G adjacent to x. Let G' be the graph formed from G by deleting the edges $xa_i, i = 1, 2, \ldots, k$. Then every vertex in $N_G(x)$ is odd in G'. Let Σ be a minimal path decomposition of the edges of G'. We shall call the following construction the Lovász construction about x with respect to $\{a_{l_1}, a_{l_2}, \ldots, a_{l_s}\}$, where $\{a_{l_1}, a_{l_2}, \ldots, a_{l_s}\} \subset \{a_1, a_2, \ldots, a_k\}$.

For each $1 \leq i \leq s$, we construct a Lovász sequence $S_{l_i} = (a_{l_i,0}, a_{l_i,1}, \ldots, a_{l_i,r_i})$. Then we define a function f that maps elements of Σ into paths and circuits that partition the edges of $G' \cup \{xa_{l_1}, xa_{l_2}, \ldots, xa_{l_s}\}$ in the same way as above. Thus $f(\Sigma)$ is a path-circuit decomposition of $G' \cup \{xa_{l_1}, xa_{l_2}, \ldots, xa_{l_s}\}$ with $|f(\Sigma)| = |\Sigma|$.

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2.2 Extension of Lovász Construction

Suppose we have performed Lovász construction about x and obtained a path-cicuit decomposition $f(\Sigma)$ of G. If q is the number of circuits formed by f, then $0 \leq q \leq \frac{1}{2}k$, where k is the number of even vertices adjacent to x.

We shall break these q circuits into edge disjoint paths in the following manner.

(I) For any three circuits C_1,C_2, and C_3 among those q circuits, let $C^* = C_1 \cup C_2 \cup C_3$. Obviously $x \in \bigcap_{i=1}^{3} V(C_i)$.

(Ia) If $V(C_i) \cap V(C_j) \setminus \{x\} = \emptyset$, for any $i \neq j$, i.e., $V(C_1), V(C_2)$, and $V(C_3)$ are pairwise disjoint except the vertex x, then we can easily break C_1, C_2, and C_3 into 4 edge disjoint paths with two paths starting at x. (See Fig.2.2 Case Ia.)

(Ib) If there exist i, j, where $i \neq j$, such that $V(C_i) \cap V(C_j) \setminus \{x\} \neq \emptyset$, let v be a vertex of $V(C^*) \setminus \{x\}$ such that

(i) v is a vertex of at least two circuits of C_1, C_2, and C_3.

(ii) subject to (i), $d_{C^*}(x, v)$ is as small as possible.

Let P be a path of C^* from x to v such that the length $|P| = d_{C^*}(x, v)$. Without loss of generality, we may assume that $P \subseteq C_1, v \in V(C_1) \cap V(C_2)$, and $N_{C_2}(v) = \{v_1, v_2\}$. We may also assume that $v_1 \neq a_3$(or $v_1 \neq b_3$). (See Fig.2.2 Case Ib.)

Let $P^* = v_1 v P x a_3$, then $C_1 \setminus P, C_2 \setminus \{v_1 v\}$, and $C_3 \setminus \{x a_3\}$ are three edge disjoint paths with $C_1 \setminus P$ and $C_3 \setminus \{x a_3\}$ starting at x. i.e., we break C_1, C_2, and C_3 into four edge disjoint paths with two paths starting at x.

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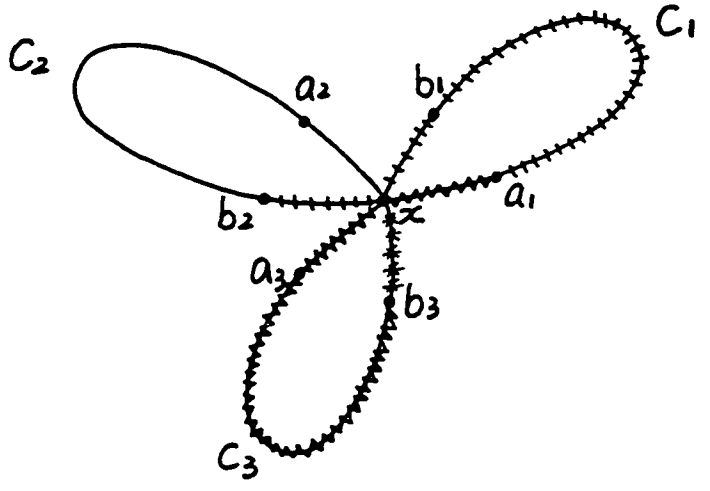
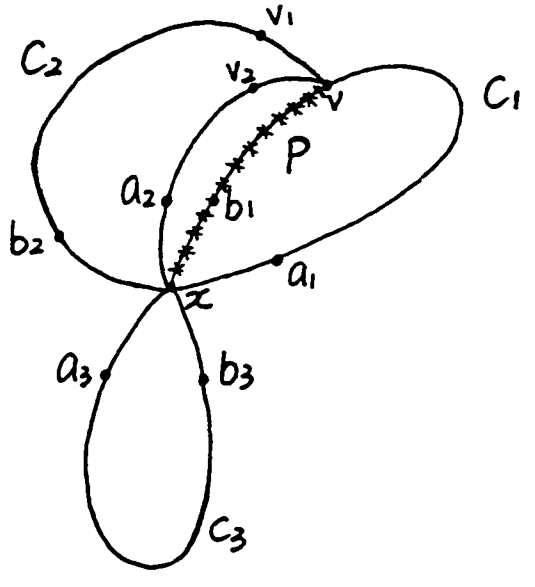


case Ia



case Ib

Figure 2.2: Break three circuits into four edge disjoint paths

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(II) If $q \geq 4$, we shall break any four circuits C_1, C_2, C_3, and C_4 into five edge disjoint paths in the following way.

Let $C^* = C_1 \cup C_2 \cup C_3 \cup C_4$, and $N_{C_i}(x) = \{a_i, b_i\}$, where $i = 1, 2, 3, 4$.

(IIa) Suppose there exists a vertex $a \in N_{C^*}(x)$ and a circuit $C \in \{C_1, C_2, C_3, C_4\}$ such that $a \notin V(C)$. Without loss of generality, we may assume that $a_1 \notin V(C_2)$.

(i) Suppose $a_2, b_2 \notin V(C_3) \cup V(C_4)$. We break the circuits C_1, C_3,and C_4 into four edge disjoint paths in the way described in (I). Among these four edge disjoint paths, there is a path P, where $P \subset C_3$(or $P \subset C_4$), such that P has x as an endpoint. (See Fig. 2.3 Case (IIa)(i).)
Since $a_2 \notin V(C_3) \cup V(C_4)$, i. e., $a_2 \notin V(P)$, then $P \cup \{xa_2\}$ is a path. And so is $C_2 \setminus \{xa_2\}$. We break C^* into five edge disjoint paths with one path starting at x.

(ii) Suppose $a_2 \in V(C_3)$ (or $a_2 \in V(C_4)$). We break the circuits C_2, C_3, and C_4 into four edge disjoint paths by introducing a path P that contains the edge xa_2. (See Fig.2.3 Case (IIa)(ii).)
Among these four edge disjoint paths, there is a path which is $C_2 \setminus \{xa_2\}$. Since $a_1 \notin V(C_2)$, then $(C_2 \setminus \{xa_2\}) \cup \{xa_1\}$ is a path. And so is $C_1 \setminus \{xa_1\}$. We break C_1, C_2, C_3, and C_4 into five edge disjoint paths with two paths starting at x.

(IIb) Suppose that $N_{C^*}(x) \subset \bigcap_{i=1}^{4} V(C_i)$. We shall find a path P of length 4 to break C_1, C_2, C_3, and C_4 into five edge disjoint paths.

Let $a_3xa_1 \subset P$ (see Fig.2.3 Case (IIb)). Since $a_1 \in V(C_2)$, let $N_{C_2}(a_1) = \{a_1^1, a_1^2\}$. Without loss of generality, we may assume that $a_1^1 \neq a_3$. Let $N_{C_4}(a_3) = \{a_3^1, a_3^2\}$. If $\{a_3^1, a_3^2\} = \{a_1, a_1^1\}$, then $a_1^2 \neq a_3$, let $P = a_1^1 a_3 x a_1 a_1^2$.

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Otherwise we may assume that $a_3^1 \notin \{a_1, a_1^1\}$ and let $P = a_3^1 a_3 x a_1 a_1^1$. Then P breaks C^* into 5 edge disjoint paths with two paths starting at x.

(III) If $q = 1$, let $\{a, b\}$ be the neighbors of x on the circuit C. Let $P = xa$, then $C \setminus \{xa\}$ and P are two edge disjoint paths of G that have x as an endpoint.

If $q = 2$, let $\{a_1, b_1\}$ be the neighbors of x on the circuit C_1, and $\{a_2, b_2\}$ be the neighbors of x on the other circuit C_2. Let $P = a_1 x a_2$, then $C_1 \setminus \{xa_1\}$, $C_2 \setminus \{xa_2\}$ and P are edge disjoint paths of G. And $C_1 \setminus \{xa_1\}$ and $C_2 \setminus \{xa_2\}$ begin with the vertex x.

Therefore, for any $q \geq 1$ we introduce $\lfloor \frac{q+3}{4} \rfloor$ edge disjoint paths to break q circuits and thus obtain a path partition of the edges of G of cardinality $|\Sigma| + \lfloor \frac{q+3}{4} \rfloor$.

We will refer to this partition as $f'(\Sigma)$. Strictly speaking, f' is not well defined since the way we break up the circuits of $f(\Sigma)$ is to some extent arbitrary; but this will pose no problems in the future since we are concerned primarily with the cardinality of $f(\Sigma)$. And also note that at least one path of $f'(\Sigma)$ has x as an endpoint.

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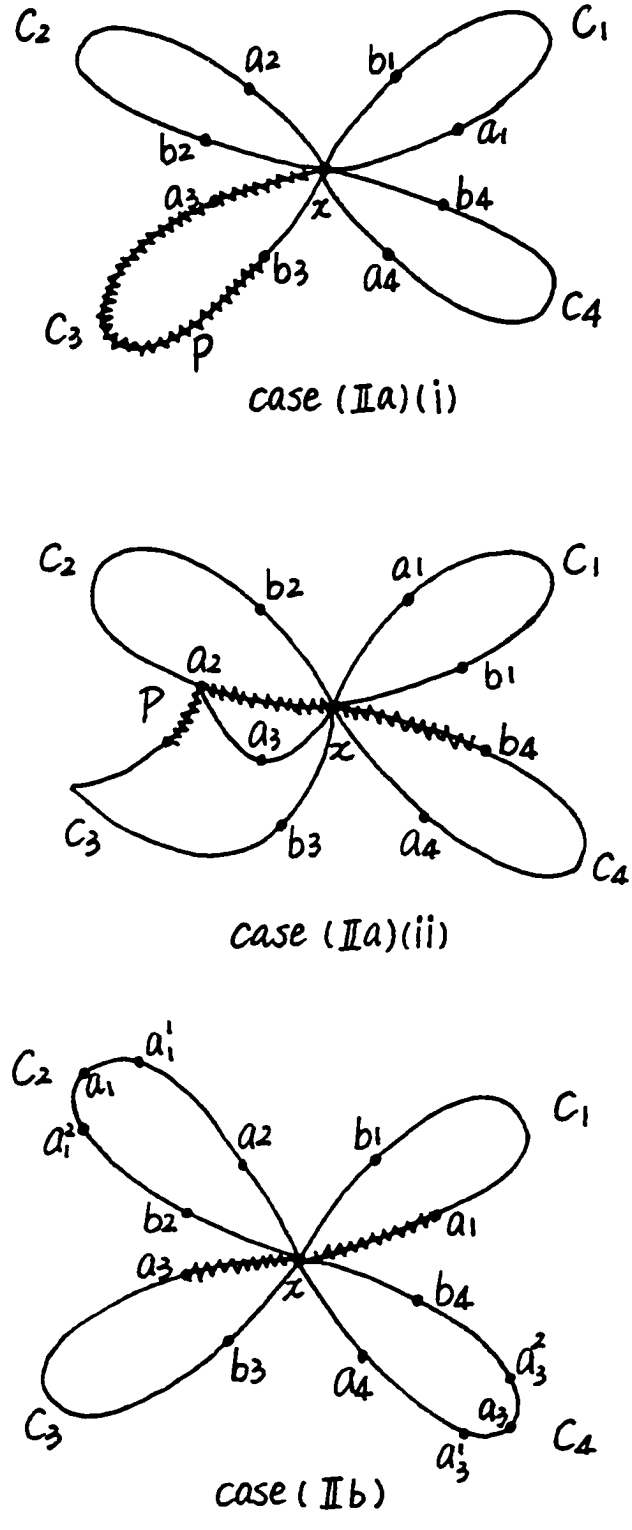

Figure 2.3: Break four circuits into five edge disjoint paths

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Chapter 3

MAIN THEOREM

The main goal of this chapter is to improve the upper bound of the path number $\wp(G)$ of a graph G. The best result so far concerning the path number $\wp(G)$ of a graph G is due to Alan Donald [6]. He proved that the path number of a graph G with n vertices is less than or equal to $\lfloor \frac{3}{4}n \rfloor$. The main result of this chapter is the following theorem, which is the main achievement of this dissertation.

Theorem 3.1 *(Main Theorem) Let G be a graph with μ vertices of odd degree and g nonisolated vertices of even degree. Then*

$$\wp(G) \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

The following corollary follows directly from the main theorem.

Corollary 3.1 *Let G be a graph with n vertices. Then*

$$\wp(G) \leq \lfloor \frac{2}{3}n \rfloor.$$

Consider a graph G such that each component is a triangle. Then $\mu = 0, g \equiv 0 \bmod 3$. So $\wp(G) = \lfloor \frac{2}{3}g \rfloor = \frac{2}{3}n$, where $n(= g)$ is the number of vertices of G. Therefore the main theorem is best possible.

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3.1 Observations on Lovász Construction

When Lovász construction produces a circuit, there must be a path $U_{i,\mu} = U_{j,\nu}$ and $a_{i,\mu}$ and $a_{j,\nu}$ are the final entries in the sequences S_i and S_j, where $i \neq j$. In this case, we say that S_i produces a circuit (with S_j). If $U_{i,\mu}$ doesn't contain x and the labeling of $U_{i,\mu}$ is unique, then in Lovász construction, $f(U_{i,\mu})$ is a path in $f(\Sigma)$ that has x as an endpoint. In this case, we call the sequence S_i produces a path starting at the vertex x. So we have the following observations:

Observation 3.1.1 If there is only one Lovász sequence, then no circuit is formed by Lovász construction. Moreover, this Lovász sequence produces a path starting at the vertex x in $f(\Sigma)$.

Observation 3.1.2 If a Lovász sequence S_i does not produce a circuit in Lovász construction, then the sequence S_i produces a path in $f(\Sigma)$ that has x as an endpoint.

Observation 3.1.3 For every vertex $v \notin \{a_1, a_2, \ldots, a_k\}$ in G, if there is a path starting at v in Σ, then there is a path starting at v in $f(\Sigma)$. And for $1 \leq i \leq k$, the number of paths starting at a_i in $f(\Sigma)$ is exactly one less than that in Σ.

The following observation on Lovász construction is made by Donald [6]. Since the parity of the degrees of the neighbors of x in G was used only to ensure that each began a path of Σ, so we have

Observation 3.1.4 Let Z be a nonempty subset of $N_G(x)$. G' is the graph obtained from G by deleting the edges joining Z to x, and Σ is a path decomposition of G'. If every neighbor of x in G begins some path of Σ, then there is a path-circuit decomposition $f(\Sigma)$ of the edges of G whose cardinality is $|\Sigma|$.

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3.2 Fundamental Lemmas

We summarize the results of Section 2.2 in the following two lemmas:

Lemma 3.2.1 *Let $\{a_1, a_2, \ldots, a_k\}$ be the set of even vertices adjacent to x in a graph G. Let $G' = G \setminus \{xa_1, xa_2, \ldots, xa_k\}$. Suppose Σ is a path decomposition of G' and we have performed Lovász construction about x to obtain a path-circuit decomposition $f(\Sigma)$ of G of cardinality $|f(\Sigma)| = |\Sigma|$. Suppose in $f(\Sigma)$ we have exactly q circuits, where $0 \leq q \leq \frac{k}{2}$, we introduce $\lfloor \frac{q+3}{4} \rfloor$ paths to break these q circuits into edge disjoint paths of G. So we have a path decomposition $f'(\Sigma)$ of G of cardinality $|f'(\Sigma)| \leq |f(\Sigma)| + \lfloor \frac{q+3}{4} \rfloor$. Therefore,*

$$\wp(G) \leq \wp(G') + \lfloor \frac{q+3}{4} \rfloor.$$

Lemma 3.2.2 *Let G be a graph with at least one even vertex, and let x be a vertex adjacent to an even vertex in G. If $f'(\Sigma)$ is a path decomposition of the edges of G obtained by performing Lovász construction and its extension about x, then at least one path in $f'(\Sigma)$ begins at x.*

Lemma 3.2.3 *Let x be an even vertex of G such that $N_e(x) = \{a_1, a_2, a_3\}$ and $|N_e(a_i)| \leq 3, i = 1, 2, 3$. Suppose that Σ is a path decomposition of $G' = G \setminus \{xa_1, xa_2, xa_3\}$ of cardinality $|\Sigma|$. Then there is a path decomposition $f'(\Sigma)$ of G such that*

(i) $|f'(\Sigma)| = |\Sigma|$, and

(ii) there is a path in $f'(\Sigma)$ starting at x.

Proof. In the graph $G' = G \setminus \{xa_1, xa_2, xa_3\}$, since every vertex v in $N_G(x)$ is an odd vertex, so there is a path in Σ starting at the vertex v.

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We perform Lovász construction about x. If Lovász construction does not produce a circuit, then we obtain a path decomposition $f'(\Sigma)$ of G of cardinality $|f'(\Sigma)| = |\Sigma|$ and there are three paths (each Lovász sequence produces one) in $f'(\Sigma)$ starting at the vertex x. We are done.

Suppose that Lovász construction does produce circuits. Since $k = 3$, there is exactly one circuit. So there is a path $U_{i,\mu} = U_{j,\nu}$ and $a_{i,\mu}$ and $a_{j,\nu}$ are the final entries in the sequences S_i and S_j, where $i \neq j$. Without loss of generality, we may assume that $i = 1$, and $j = 2$.

Now instead of performing Lovász construction about the vertex x, we perform Lovász construction about x with respect to $\{a_2, a_3\}$. Then S_2 and S_3 are the only Lovász sequences we have and they do not produce a circuit. So $f(\Sigma)$ is a path decomposition of $G \setminus \{xa_1\}$ of cardinality $|f(\Sigma)| = |\Sigma|$. By Observation 3.1.2, there are two paths (each Lovász sequence produce one) in $f(\Sigma)$ starting at the vertex x. Moreover, since x is odd in G', so there is a path in Σ, hence in $f(\Sigma)$, starting at x. Therefore, there are three different paths P_1, P_2, and P_3 in $f(\Sigma)$ starting at the vertex x.

Since $|N_e(a_1))| \leq 3$, let $N_e(a_1) = \{x, b_1, b_2\}$, where $x = b_1 = b_2$ if $|N_e(a_1))| = 1$, and $x = b_2 \neq b_1$ if $|N_e(a_1))| = 2$.

Note that if $b_i \in \{a_2, a_3\}$, then by Observation 3.1.3, there may be no path in $f(\Sigma)$ starting at the vertex b_i, where $i = 1, 2$. But for each vertex $w \in N_G(a_1)$ but $\notin \{b_1, b_2\}$, there is a path in $f(\Sigma)$ starting at w.

Now for the vertex x, we construct three Lovász sequences S_i, each with $U_{i,0} = P_i$ as a starting path and $x_{i,0} = x$ as an end point of $U_{i,0}$, where $i = 1, 2, 3$. Suppose $x_{i,\mu}$ is defined. If a_1 lies on the path $U_{i,\mu}$, we trace the path from $x_{i,\mu}$ to a_1 and let $x_{i,\mu+1}$ be the last vertex on the path before a_1. The sequence ends either when $x_{i,\mu} = b_1$ or b_2 or when the path $U_{i,u}$ does not contain the vertex a_1.

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We claim that no two sequences of S_i, $i = 1, 2, 3$, share a common entry other than the vertex x.

Suppose $x_{i,\mu} = x_{j,\nu} \neq x$, where $\mu \leq \nu$, is a common entry of S_i and S_j, where $i \neq j$. Then $x_{i,\mu}a_1 = x_{j,\nu}a_1 \in U_{i,\mu-1} \cap U_{j,\nu-1}$. Since the paths in $f(\Sigma)$ are edge disjoint, so $U_{i,\mu-1} = U_{j,\nu-1}$. If we continue this tracing process, we have $P_i = U_{i,0} = U_{j,\nu-\mu}$. If $\mu = \nu$, then $P_i = P_j$, where $i \neq j$. If $\mu \leq \nu$, then $P_i = U_{j,\nu-\mu}$. This means that the edge xa_1 lies on $U_{j,\nu-\mu-1}$. Both are impossible.

Therefore, among these three sequences there is at least one sequence such that neither b_1 nor b_2 is an entry of the sequence. So we perform Lovász construction about the vertex a_1 with respect to $\{x\}$ with this sequence. We obtain a path decomposition $f'(\Sigma)$ of G of cardinality $|f'(\Sigma)| = |\Sigma|$ and there is at least one path in $f'(\Sigma)$ starting at x.

3.3 The Proof of the Main Theorem

In this section, we shall present a complete proof of the main theorem. The key idea in the proof is developed from Donald [2], which relies heavily on Lovász construction.

For the sake of the completeness of this section, we restate the main theorem in the following:

Main Theorem Let G be a graph with μ vertices of odd degree and g nonisolated vertices of even degree. Then

$$\wp(G) \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

Proof. The proof is, as in Lovász theorem, by induction on $\lambda(G) = 2m - n$, where m is the number of edges in G and n is the number of nonisolated vertices.

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If $\lambda(G) = 0$, i. e. $2m = n$, then G is a matching, possibly with some isolated vertices. The theorem is straightforward.

If $g = 0$, Corollary 1.1.1 gives the result immediately. So we may assume that G contains at least one nonisolated even vertex.

Suppose the theorem is true for any graph G' with $\lambda(G') < \lambda(G)$.

<u>Case 1</u> : G contains at least one pair of adjacent even vertices.

Let x be one such vertex and let $a_1, a_2, \ldots, a_k$ be the even vertices adjacent to x, where $k \geq 1$. Let

$$G' = G \setminus \{xa_1, xa_2, \ldots, xa_k\}.$$

Then $m' = m - k, n' \geq n - 1$ (If $|N_G(x)| = k$, then $n' = n - 1$. Otherwise, $n' = n$.), where m' is the number of edges and n' the number of nonisolated vertices of G'. So

$$\begin{aligned} \lambda(G') &= 2m' - n' \leq 2(m-k) - (n-1) \\ &= 2m - n - 2k + 1 \\ &< 2m - n = \lambda(G). \end{aligned}$$

By the induction hypothesis, we have

$$\wp(G') \leq \frac{1}{2}\mu' + \lfloor \frac{2}{3} g' \rfloor,$$

where μ' is the number of odd vertices and g' the number of nonisolated even vertices of G'.

Let Σ be a path decomposition of G' of cardinality $\wp(G')$. We perform Lovász construction about the vertex x. If q circuits are produced, then by Lemma 3.2.1, we have

$$\begin{aligned} \wp(G) &\leq \wp(G') + \lfloor \tfrac{q+3}{4} \rfloor \\ &\leq \tfrac{1}{2}\mu' + \lfloor \tfrac{2}{3} g' \rfloor + \lfloor \tfrac{q+3}{4} \rfloor \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*) \end{aligned}$$

<u>Subcase 1a</u> : k is odd.

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Since k is odd, so $\mu' = \mu + k + 1, g' = g - k - 1$, and $q \leq \frac{k-1}{2}$.

Thus Eq.(*) gives:

$$\begin{aligned}\wp(G) &\leq \tfrac{1}{2}(\mu + k + 1) + \lfloor \tfrac{2}{3}(g - k - 1) \rfloor + \lfloor \tfrac{q+3}{4} \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{1}{2}(k+1) - \lfloor \tfrac{2}{3}(k+1) \rfloor + \lfloor \tfrac{k+5}{8} \rfloor.\end{aligned}$$

If $k \equiv 1 \bmod 8$, then $\lfloor \frac{k+5}{8} \rfloor = \lfloor \frac{k-1}{8} \rfloor$.

$$\begin{aligned}\wp(G) &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{12k+12}{24} - \lfloor \tfrac{16k+16}{24} \rfloor + \lfloor \tfrac{3k-3}{24} \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor.\end{aligned}$$

If $k \equiv 3 \bmod 8$, and $k \neq 3$, then

$$\begin{aligned}\wp(G) &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{12k+12}{24} - \lfloor \tfrac{16k+16}{24} \rfloor + \lfloor \tfrac{3k+15}{24} \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor.\end{aligned}$$

If $k \equiv 5 \bmod 8$, then $\lfloor \frac{k+5}{8} \rfloor = \lfloor \frac{k+3}{8} \rfloor$.

$$\begin{aligned}\wp(G) &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{12k+12}{24} - \lfloor \tfrac{16k+16}{24} \rfloor + \lfloor \tfrac{3k+9}{24} \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor.\end{aligned}$$

If $k \equiv 7 \bmod 8$, then $\lfloor \frac{k+5}{8} \rfloor = \lfloor \frac{k+1}{8} \rfloor$.

$$\begin{aligned}\wp(G) &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{12k+12}{24} - \lfloor \tfrac{16k+16}{24} \rfloor + \lfloor \tfrac{3k+3}{24} \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor.\end{aligned}$$

<u>Subcase 1b</u> : k is even.

Since k is even, so $u' = \mu + k, g' = g - k$, and $q \leq \frac{k}{2}$.

Thus Eq.(*) gives:

$$\begin{aligned}\wp(G) &\leq \tfrac{1}{2}(\mu + k) + \lfloor \tfrac{2}{3}(g - k) \rfloor + \lfloor \tfrac{q+3}{4} \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{1}{2}k - \lfloor \tfrac{2}{3}k \rfloor + \lfloor \tfrac{k+6}{8} \rfloor.\end{aligned}$$

If $k \equiv 0 \bmod 8$, then $\lfloor \frac{k+6}{8} \rfloor = \lfloor \frac{k}{8} \rfloor$.

$$\begin{aligned}\wp(G) &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{1}{2}k - \lfloor \tfrac{2}{3}k \rfloor + \lfloor \tfrac{1}{8}k \rfloor \\ &= \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{12k}{24} - \lfloor \tfrac{16k}{24} \rfloor + \lfloor \tfrac{3k}{24} \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor.\end{aligned}$$

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If $k \equiv 4 \bmod 8$, and $k \neq 4$, then $\lfloor \frac{k+6}{8} \rfloor = \lfloor \frac{k+4}{8} \rfloor$.

$$\begin{aligned} \varphi(G) &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{1}{2}k - \lfloor \tfrac{2}{3}k \rfloor + \lfloor \tfrac{k+4}{8} \rfloor \\ &= \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{12k}{24} - \lfloor \tfrac{16k}{24} \rfloor + \lfloor \tfrac{3k+12}{24} \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor. \end{aligned}$$

If $k \equiv 6 \bmod 8$, then $\lfloor \frac{k+6}{8} \rfloor = \lfloor \frac{k+2}{8} \rfloor$.

$$\begin{aligned} \varphi(G) &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{1}{2}k - \lfloor \tfrac{2}{3}k \rfloor + \lfloor \tfrac{k+2}{8} \rfloor \\ &= \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor + \tfrac{12k}{24} - \lfloor \tfrac{16k}{24} \rfloor + \lfloor \tfrac{3k+6}{24} \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor. \end{aligned}$$

If $k \equiv 2 \bmod 8$, we adjoin to G one vertex y and the edge $a_k y$ to obtain the graph G_1. We claim that $\varphi(G) \leq \varphi(G_1)$, for if Σ is a path decomposition of G_1, then the edge $a_k y$ is the initial edge of a path of Σ. We can remove this edge, reducing the cardinality of the partition by 1 if $a_k y$ is itself a path and by 0 if $a_k y$ is not an entire path.

We now show that $\varphi(G_1) \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor$.

As before, we delete $xa_i, i = 1, 2, \ldots, k-1$, of the graph G_1 to obtain the graph G_1'. Since $k \equiv 2 \bmod 8, k-1 \geq 1$, so we delete at least one edge and do not isolate the even vertex x.

Let m_1' (respectively, m_1) be the number of edges and n_1' (respectively, n_1) the number of nonisolated vertices of G_1' (respectively, G_1).

Then

$$m_1' = m_1 - (k-1) = (m+1) - (k-1) = m - k + 2, \; n_1' = n_1 = n + 1,$$

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and

$$\begin{aligned}\lambda(G_1') &= 2m_1' - n_1' = 2(m-k+2) - (n+1)\\ &= 2m - n - 2k + 3\\ &< 2m - n = \lambda(G).\end{aligned}$$

By the induction hypothesis, we have $\varphi(G_1') \leq \frac{1}{2}\mu_1' + \lfloor \frac{2}{3} g_1' \rfloor$, where μ_1' is the number of odd vertices and g_1' the number of nonisolated even vertices of G_1'.

Since

$$\mu_1' = \mu + k + 2 \text{ and } g_1' = g - k - 1,$$

Thus Eq.(*) gives:

$$\begin{aligned}\varphi(G_1) &\leq \tfrac{1}{2}\mu_1' + \lfloor \tfrac{2}{3} g_1' \rfloor + \lfloor \tfrac{q+3}{4} \rfloor \text{ (where } q \leq \tfrac{k-2}{2})\\ &\leq \tfrac{1}{2}\mu_1' + \lfloor \tfrac{2}{3} g_1' \rfloor + \lfloor \tfrac{k+4}{8} \rfloor\\ &= \tfrac{1}{2}(\mu + k + 2) + \lfloor \tfrac{2}{3}(g - k - 1) \rfloor + \lfloor \tfrac{k-2}{8} \rfloor \text{ (since } k \equiv 2 \bmod 8)\\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3} g + \tfrac{12k+24-16k-16+3k-6}{24} \rfloor\\ &= \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3} g + \tfrac{2-k}{24} \rfloor\\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3} g \rfloor.\end{aligned}$$

<u>Subcase 1c</u> : $k = 3$.

We may assume that for every even vertex $v \in V(G)$, if $N_e(v) \neq \emptyset$, then $|N_e(v)| = 3$ or 4. For if there exists an even vertex v such that $N_e(v) \neq \emptyset$ and $|N_e(v)| \neq 3$ or 4, then choose this vertex v as the vertex x in the previous proof. We are done.

Since $k = 3$, let $N_e(x) = \{a_1, a_2, a_3\}$.

(i) $|N_e(a_i)| = 3$ for $i = 1, 2, 3$.

Let $G' = G \setminus \{xa_1, xa_2, xa_3\}$, then $\mu' = \mu + 4, g' = g - 4$.

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By the induction hypothesis, we have

$$\begin{aligned}\wp(G') &\leq \tfrac{1}{2}\mu' + \lfloor \tfrac{2}{3}g' \rfloor \\ &= \tfrac{1}{2}(\mu+4) + \lfloor \tfrac{2}{3}(g-4) \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor.\end{aligned}$$

Let Σ be a path decomposition of G' of cardinality $|\Sigma| = \wp(G')$. Since $|N_e(x)| = 3$, and $|N_e(a_i)| = 3$, for $i = 1, 2, 3$, by Lemma 3.2.3, we have a path decomposition $f'(\Sigma)$ of G such that $|f'(\Sigma)| = |\Sigma|$ and there is a path in $f'(\Sigma)$ starting at the vertex x.

So

$$\wp(G) \leq \wp(G') \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

(ii) There exists a vertex $a \in \{a_1, a_2, a_3\}$ such that $|N_e(a)| = 4$.

Without loss of generality, we may assume that $a = a_1$.

Let

$$N_e(a_1) = \{x, b_1, b_2, b_3\}.$$

(a) $a_2, a_3 \notin \{b_1, b_2, b_3\}$.

Let

$$G' = G \setminus \{xa_1, xa_2, xa_3, a_1b_1, a_1b_2, a_1b_3\},$$

then

$$\mu' = \mu + 6, g' \leq g - 6. \text{ (See Fig. 3.1 .)}$$

(If $|N_G(a_1)| = 4$, then $g' = g - 7$. Otherwise, $g' = g - 6$.)

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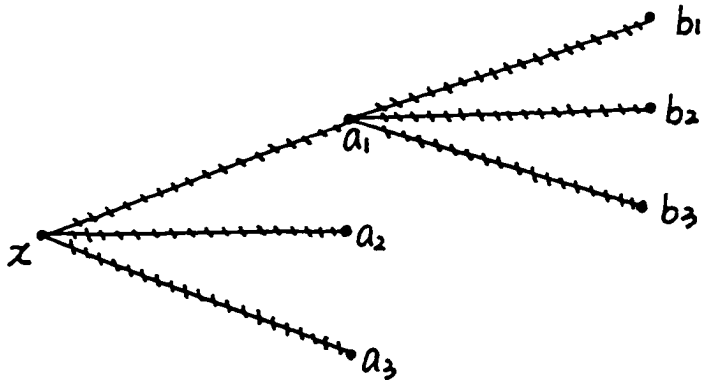

Figure 3.1: The relationship between graphs G and G'

By the induction hypothesis, we have

$$\begin{aligned}\wp(G') &\leq \tfrac{1}{2}\mu' + \lfloor \tfrac{2}{3}g' \rfloor \\ &\leq \tfrac{1}{2}(\mu+6) + \lfloor \tfrac{2}{3}(g-6) \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor - 1.\end{aligned}$$

Let Σ be a path decomposition of G' of cardinality $|\Sigma| = \wp(G')$. Since every neighbor of a_1 in G is an odd vertex in G', so we may perform Lovász construction about the vertex a_1 with respect to $\{b_1, b_2, b_3\}$. We obtain a path-circuit decomposition $f(\Sigma)$ of $G' \cup \{a_1b_1, a_1b_2, a_1b_3\}$ of cardinality $|f(\Sigma)| = |\Sigma|$ with at most one circuit and at least one path starting at a_1. If there is a circuit C_1 in $f(\Sigma)$, let $\{y_1, y_2\}$ be the neighbors of a_1 on C_1. Since $b_1, b_2,$ and $b_3 \notin N_G(x)$, by Observation 3.1.3, there is a path in $f(\Sigma)$ starting at each neighbor of x in G. After performing Lovász construction about the vertex x with respect to $\{a_2, a_3\}$, we obtain a path-circuit decomposition $f^*(\Sigma)$ of $G \setminus \{xa_1\}$ of cardinality $|f^*(\Sigma)| = |f(\Sigma)| = |\Sigma|$ with at most two circuits (including the circuit C_1, if it exists). If there

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are two circuits in $f^*(\Sigma)$, let $\{z_1, z_2\}$ be the neighbors of x on the second circuit C_2.

If C_1 exists and C_2 does not exist,

let

$$P^* = xa_1y_1,$$

then

$$(f^*(\Sigma) \setminus C_1) \cup P^* \cup (C_1 \setminus \{a_1y_1\})$$

is a path decomposition of G with

$$\wp(G) \leq \wp(G') + 1 \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

If both C_1 and C_2 exist, we may assume that $y_1 \neq z_1$.

Let

$$P^* = y_1a_1xz_1,$$

then

$$(f^*(\Sigma) \setminus \{C_1, C_2\}) \cup P^* \cup (C_1 \setminus \{a_1y_1\}) \cup (C_2 \setminus \{xz_1\})$$

is a path decomposition of G with

$$\wp(G) \leq \wp(G') + 1 \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

If both C_1 and C_2 do not exist,

let

$$P^* = xa_1,$$

then

$$f^*(\Sigma) \cup P^*$$

is a path decomposition of G with

$$\wp(G) \leq \wp(G') + 1 \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

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(b) a_2 (ora_3) $\in \{b_1, b_2, b_3\}$, but not both.

Without loss of generality, we may assume that $a_2 = b_3$.

Let

$$G' = G \setminus \{xa_1, xa_2, xa_3, a_1b_1, a_1b_2\},$$

then

$$\mu' = \mu + 6, g' = g - 6. \text{ (See Fig. 3.2 .)}$$

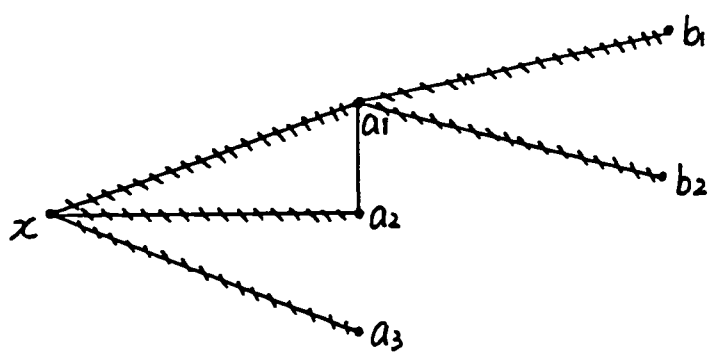

Figure 3.2: The relationship between graphs G and G'

By the induction hypothesis, we have

$$\begin{aligned} \wp(G') &\leq \tfrac{1}{2}\mu' + \lfloor \tfrac{2}{3}g' \rfloor \\ &= \tfrac{1}{2}(\mu + 6) + \lfloor \tfrac{2}{3}(g - 6) \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor - 1. \end{aligned}$$

Let Σ denote a path decomposition of G' of cardinality $|\Sigma| = |\wp(G')|$. For the vertex b_1, we construct a Lovász sequence and perform Lovász construction about a_1 with respect to $\{b_1\}$. We obtain a path decomposition $f(\Sigma)$ of $G' \cup \{a_1b_1\}$ of cardinality $|f(\Sigma)| = |\Sigma|$.

Let

$$P^* = b_2a_1xa_2,$$

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then

$$f(\Sigma) \cup P^*$$

is a path decomposition of $G \setminus \{xa_3\}$ and

$$\begin{aligned} |f(\Sigma) \cup P^*| &= |\Sigma| + 1 \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor - 1 + 1 \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor. \end{aligned}$$

Since every neighbor of x in G starts some path in $f(\Sigma) \cup P^*$, we perform Lovász construction about x with respect to $\{a_3\}$. By Observation 3.1.4, there is a path-circuit decomposition of the edges of G of cardinality $|f(\Sigma) \cup P^*|$. Since there is only one Lovász sequence, by Observation 3.1.1, this path-circuit decomposition is in fact a path decomposition of G. Thus

$$\varphi(G) \leq |f(\Sigma) \cup P^*| \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

(c) $a_2, a_3 \in \{b_1, b_2, b_3\}$.

Without loss of the generality, we may assume that $\{a_2, a_3\} = \{b_2, b_3\}$. We adjoin to G one vertex y and the edge a_3y to obtain the graph G_1. Obviously, $\varphi(G) \leq \varphi(G_1)$, $\mu_1 = \mu + 2$, and $g_1 = g - 1$.

Let

$$G_1' = G_1 \setminus \{xa_2, a_1b_1\},$$

then

$$\mu_1' = \mu_1 + 4 = \mu + 6, g_1' = g_1 - 4 = g - 5. \text{ (See Fig. 3.3 .)}$$

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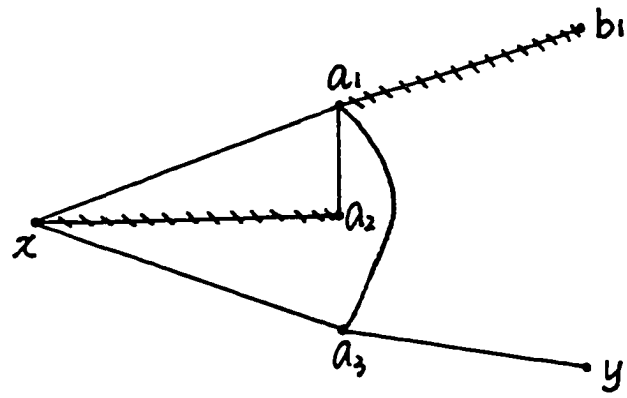

Figure 3.3: The relationship between graphs G and G'

By the induction hypothesis, we have

$$\begin{aligned} \wp(G_1') &\leq \tfrac{1}{2}\mu' + \lfloor \tfrac{2}{3}g' \rfloor \\ &= \tfrac{1}{2}(\mu+6) + \lfloor \tfrac{2}{3}(g-5) \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor. \end{aligned}$$

Let Σ_1 denote a path decomposition of G_1' of cardinality $|\Sigma_1| = \wp(G_1')$. Since each neighbor of a_1 in G is an odd vertex in G_1', we perform Lovász construction about a_1 with respect to $\{b_1\}$ to obtain a path decomposition $f(\Sigma_1)$ of $G_1 \setminus \{xa_2\}$ of cardinality $|f(\Sigma_1)| = |\Sigma_1| = \wp(G_1')$.

In $f(\Sigma_1)$, there is a path starting at each neighbor of x in G. We perform Lovász construction about x with respect to $\{a_2\}$, we obtain a path decomposition $f'(\Sigma_1)$ of G_1 with

$$\begin{aligned} \wp(G_1) &\leq |f'(\Sigma_1)| = |f(\Sigma_1)| = |\Sigma_1| \\ &= \wp(G_1') \leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor. \end{aligned}$$

Remove the edge a_3y from $f'(\Sigma_1)$, we obtain a path decomposition $f'(\Sigma)$

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of G such that

$$\wp(G) \leq \wp(G_1) \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

<u>Subcase 1d</u> : $k = 4$.

We may assume that for every even vertex $v \in V(G)$, if $N_e(v) \neq \emptyset$, then $|N_e(v)| = 4$.

Let

$$N_e(x) = \{a_1, a_2, a_3, a_4\}$$

and

$$N_e(a_1) = \{x, b_1, b_2, b_3\}.$$

(i) $a_2, a_3, a_4 \notin \{b_1, b_2, b_3\}$.

Let

$$G' = G \setminus \{xa_1, xa_2, xa_3, xa_4, a_1b_1, a_1b_2, a_1b_3\},$$

then

$$\mu' = \mu + 6, g' \leq g - 6. \text{ (See Fig. 3.4 .)}$$

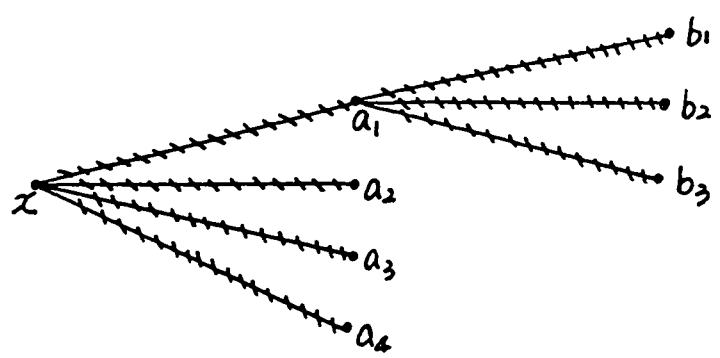

Figure 3.4: The relationship between graphs G and G'

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By the induction hypothesis, we have

$$\begin{aligned}\varphi(G') &\leq \tfrac{1}{2}\mu' + \lfloor \tfrac{2}{3}g' \rfloor \\ &= \tfrac{1}{2}(\mu+6) + \lfloor \tfrac{2}{3}(g-6) \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor - 1.\end{aligned}$$

Let Σ be a path decomposition of G' of cardinality $|\Sigma| = \varphi(G')$. We perform Lovász construction about a_1 with respect to $\{b_1, b_2, b_3\}$. It produces a path-circuit decomposition $f(\Sigma)$ of $G' \cup \{a_1b_1, a_1b_2, a_1b_3\}$ of cardinality $|f(\Sigma)| = |\Sigma|$ with at most one circuit and at least one path starting at a_1.

Note that there is a path in $f(\Sigma)$ starting at each neighbor of x in G. So we may perform Lovász construction about x with respect to $\{a_2, a_3, a_4\}$. And we have a path-circuit decomposition $f^*(\Sigma)$ of $G \setminus \{xa_1\}$ of cardinality $|f^*(\Sigma)| = |f(\Sigma)| = |\Sigma|$ with at most two circuits.

Similar to subcase 1c (ii)(a), we introduce an extra path to break the circuits (if any) of $f^*(\Sigma)$. Therefore we obtain a path decomposition of G with

$$\varphi(G) \leq |f^*(\Sigma)| + 1 = \varphi(G') + 1 \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

(ii) Exactly one of $\{a_2, a_3, a_4\}$ is in $\{b_1, b_2, b_3\}$.

Without loss of generality, we may assume that $a_2 \in \{b_1, b_2, b_3\}$. We may further assume that $a_2 = b_3$. Let

$$G' = G \setminus \{xa_1, xa_2, xa_3, xa_4, a_1b_1, a_1b_2\},$$

then

$$\mu' = \mu + 6, g' = g - 6. \text{ (See Fig. 3.5 .)}$$

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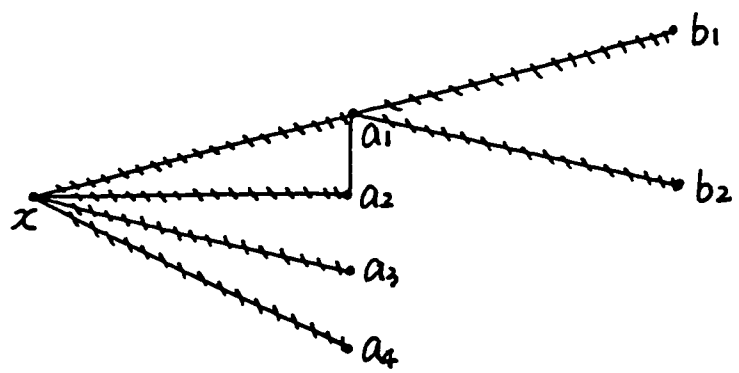

Figure 3.5: The relationship between graphs G and G'

By the induction hypothesis, we have

$$\begin{aligned} \wp(G') &\leq \tfrac{1}{2}\mu' + \lfloor \tfrac{2}{3}g' \rfloor \\ &= \tfrac{1}{2}(\mu+6) + \lfloor \tfrac{2}{3}(g-6) \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor - 1. \end{aligned}$$

Let Σ be a path decomposition of G' of cardinality $|\Sigma| = \wp(G')$. We perform Lovász construction about a_1 with respect to $\{b_1, b_2\}$. It produces a path-circuit decomposition $f(\Sigma)$ of $G' \cup \{a_1b_1, a_1b_2\}$ of cardinality $|f(\Sigma)| = |\Sigma|$ with at most one circuit and at least one path starting at a_1 (since a_1 is odd in G').

Note that there is a path in $f(\Sigma)$ starting at each neighbor of x in G. So we may perform Lovász construction about x with respect to $\{a_2, a_3, a_4\}$. And we have a path-circuit decomposition $f^*(\Sigma)$ of $G \setminus \{xa_1\}$ of cardinality $|f^*(\Sigma)| = |f(\Sigma)| = |\Sigma|$ with at most two circuits.

Similar to subcase 1c (ii)(a), we introduce an extra path to break the circuits,

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if any, of $f^*(\Sigma)$. Therefore we obtain a path decomposition of G with

$$\wp(G) \leq |f^*(\Sigma)| + 1 = \wp(G') + 1 \leq \frac{1}{2}\mu + \lfloor \frac{2}{3} g \rfloor.$$

(iii) Exactly two of $\{a_2, a_3, a_4\}$ is in $\{b_1, b_2, b_3\}$.

Without loss of generality, we may assume that $\{a_2, a_3\} \in \{b_1, b_2, b_3\}$. We may further assume that $\{a_2, a_3\} = \{b_2, b_3\}$.

We adjoin to G two vertices y,z and the edges a_3y,a_4z to obtain the graph G_1. Then $\wp(G) \leq \wp(G_1)$, $\mu_1 = \mu + 4$, and $g_1 = g - 2$.

Let

$$G_1' = G_1 \setminus \{xa_2, a_1b_1\},$$

then

$$m_1' = m,\ n_1' = n + 2,$$

and

$$\mu_1' = \mu_1 + 4 = \mu + 8,\ g_1' = g_1 - 4 = g - 6,$$

where m_1' is the number of edges, n_1' the number of nonisolated vertices, μ_1' the number of odd vertices, and g_1' the number of nonisolated even vertices of G_1'. (See Fig. 3.6 .)

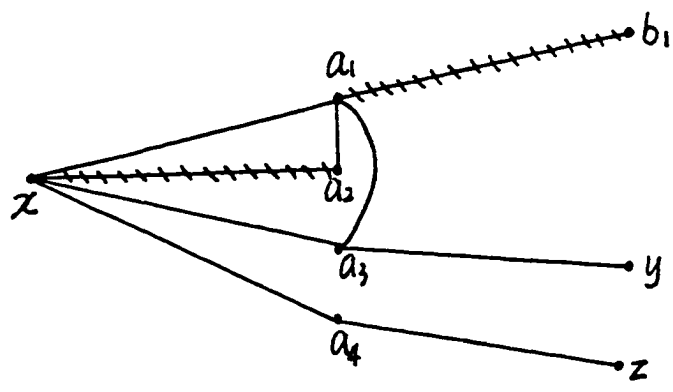

Figure 3.6: The relationship between graphs G and G'

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Since

$$\begin{aligned}\lambda(G_1') &= 2m_1' - n_1' = 2m - (n+2)\\ &< 2m - n = \lambda(G),\end{aligned}$$

by the induction hypothesis, we have

$$\begin{aligned}\wp(G_1') &\leq \tfrac{1}{2}\mu_1' + \lfloor \tfrac{2}{3}g_1' \rfloor\\ &= \tfrac{1}{2}(\mu+8) + \lfloor \tfrac{2}{3}(g-6) \rfloor\\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor.\end{aligned}$$

Let Σ_1 denote a path decomposition of G_1' of cardinality $|\Sigma_1| = \wp(G_1')$. Since each neighbor of a_1 in G is an odd vertex in G_1', we perform Lovász construction about a_1 with respect to $\{b_1\}$ and obtain a path decomposition $f(\Sigma_1)$ of $G_1 \setminus \{xa_2\}$ of cardinality $|f(\Sigma_1)| = |\Sigma_1| = \wp(G_1')$.

In $f(\Sigma_1)$, there is a path starting at each neighbor of x in G. We perform Lovász construction about x with respect to $\{a_2\}$, we obtain a path decomposition $f'(\Sigma_1)$ of G_1 with

$$\begin{aligned}\wp(G_1) &\leq |f'(\Sigma_1)| = |f(\Sigma_1)|\\ &= \wp(G_1') \leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor.\end{aligned}$$

Remove the edge a_3y and a_4z from $f'(\Sigma_1)$, we obtain a path decomposition $f'(\Sigma)$ of G such that

$$\wp(G) \leq \wp(G_1) \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

(iv) We may assume that for each a_i, $N_e(a_i) = \{x, a_1, a_2, a_3, a_4\} \setminus \{a_i\}, i = 1, 2, 3, 4$.

We adjoin to G a vertex z and the edge a_4z to obtain the graph G_1. Then $\wp(G) \leq \wp(G_1), \mu_1 = \mu + 2$, and $g_1 = g - 1$.

Let

$$G_1' = G_1 \setminus \{xa_1, xa_2, xa_3\},$$

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then

$$\mu_1' = \mu_1 + 4 = \mu + 6, g_1' = g_1 - 4 = g - 5. \text{ (See Fig. 3.7 .)}$$

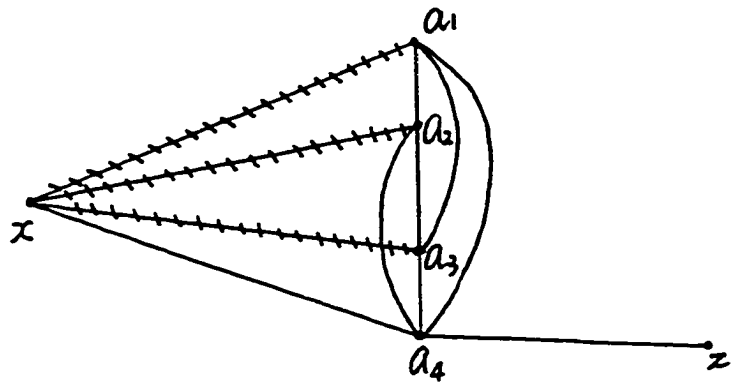

Figure 3.7: The relationship between graphs G_1 and G_1'

By the induction hypothesis, we have

$$\begin{aligned} \wp(G_1') &\leq \tfrac{1}{2}\mu_1' + \lfloor \tfrac{2}{3} g_1' \rfloor \\ &= \tfrac{1}{2}(\mu + 6) + \lfloor \tfrac{2}{3}(g - 5) \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3} g \rfloor. \end{aligned}$$

Since in the graph G_1, $|N_e(x)| = 3, |N_e(a_i)| = 3$, for $i = 1, 2, 3$, similar to Case 1c (i), we have a path decomposition $f'(\Sigma_1)$ of G_1 with $\wp(G_1) \leq \wp(G_1') \leq \frac{1}{2}\mu + \lfloor \frac{2}{3} g \rfloor$.

Remove the edge $a_4 y$ in $f'(\Sigma_1)$, we obtain a path decomposition $f'(\Sigma)$ of G with $\wp(G) \leq \wp(G_1) \leq \frac{1}{2}\mu + \lfloor \frac{2}{3} g \rfloor$.

<u>Case 2</u> : G contains no pairs of adjacent even vertices and also there exists a nonisolated even vertex y such that for any other even vertex z, $d_G(y, z) \geq 3$. (This case also includes, trivially, the case in which $g = 1$).

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Let x be an odd vertex adjacent to y.

Let

$$G' = G \setminus \{xy\}.$$

Then

$$\mu' = \mu, g' = g.$$

By the induction hypothesis, we have

$$\begin{aligned} \wp(G') &\leq \tfrac{1}{2}\mu' + \lfloor \tfrac{2}{3}g' \rfloor \\ &= \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor. \end{aligned}$$

Since x is adjacent to no other even vertex in G, i. e. every neighbor of x in G is odd in G'. We perform Lovász construction about the vertex x. Since there is only one Lovász sequence, by Observation 3.1.1, no circuit is formed. So we have a path decomposition of G with

$$\wp(G) = \wp(G') \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

<u>Case 3</u> : If neither case 1 nor case 2 holds, there is a pair of even vertices x and z such that $d_G(x, z) = 2$ and neither x nor z is adjacent to any even vertex. So there is an odd vertex y such that xy and yz are edges of G.

Delete the edge xy to obtain the graph G_1 with μ_1 odd vertices and g_1 nonisolated even vertices. Now y is even and x is odd in G_1. (Clearly, $\mu_1 = \mu$ and $g_1 = g$.) Moreover, x is adjacent to only odd vertices in G_1; and G_1 contains a pair of adjacent even vertices y and z.

Note that for every vertex $v \in N_e(y)$ in G_1, $N_e(v) = y$, i. e. v has only one even neighbor y. Let k be the number of even neighbors of y in G_1. If $k \neq 3, 4$,

let

$$G_1' = G_1 \setminus \bigcup_{i=1}^{k} yw_i,$$

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where

$$w_i \in N_e(y).$$

As in subcase 1a and subcase 1b, we complete Lovász construction and its extension about the vertex y to obtain a path decomposition $f'(\Sigma_1)$ of G_1 with

$$\varphi(G_1) \leq \frac{1}{2}\mu_1 + \lfloor \frac{2}{3}g_1 \rfloor = \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor.$$

Furthermore, there is at least one path in $f'(\Sigma_1)$ that has y as an endpoint.

If $k = 3$, let

$$N_e(y) = \{a_1 = z, a_2, a_3\}$$

and

$$G_1' = G_1 \setminus \{ya_1, ya_2, ya_3\}.$$

Then

$$\mu_1' = \mu_1 + 4 = \mu + 4,\ g_1' = g_1 - 4 = g - 4.$$

By the induction hypothesis, there is a path decomposition Σ_1 of G_1' such that

$$\begin{aligned} \varphi(G_1') &= |\Sigma_1| \\ &\leq \tfrac{1}{2}\mu_1' + \lfloor \tfrac{2}{3}g_1' \rfloor \\ &\leq \tfrac{1}{2}(\mu + 4) + \lfloor \tfrac{2}{3}(g - 4) \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3}g \rfloor. \end{aligned}$$

Since $|N_e(y)| = 3$, and $|N_e(a_i)| = 1$, for $i = 1, 2, 3$, by Lemma 3.2.3, we have a path decomposition $f'(\Sigma_1)$ of G_1 of cardinality $|f'(\Sigma_1)| = |\Sigma_1| \leq \frac{1}{2}\mu + \lfloor \frac{2}{3}g \rfloor$. And there is at least one path in $f'(\Sigma_1)$ that has y as an endpoint.

If $k = 4$, then adjoin to G_1 one vertex w and the edge wz to obtain the graph G_2. Let $\{a_1, a_2, a_3\}$ be the even neighbors of y in G_2. Then $\mu_2 = \mu_1 + 2, g_2 = g_1 - 1$. Let

$$G_2' = G_2 \setminus \{ya_1, ya_2, ya_3\},$$

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then

$$\mu_2' = \mu_2 + 4 = \mu_1 + 6 = \mu + 6,\ g_2' = g_2 - 4 = g_1 - 5 = g - 5.$$

By the induction hypothesis, there is a path decomposition Σ_2 of G_2' such that

$$\begin{aligned} \wp(G_2') &= |\Sigma_2| \\ &\leq \tfrac{1}{2}\mu_2' + \lfloor \tfrac{2}{3} g_2' \rfloor \\ &= \tfrac{1}{2}(\mu + 6) + \lfloor \tfrac{2}{3}(g - 5) \rfloor \\ &\leq \tfrac{1}{2}\mu + \lfloor \tfrac{2}{3} g \rfloor. \end{aligned}$$

In the graph G_2, $|N_e(y)| = 3, |N_e(a_i)| = 1$, for $i = 1, 2, 3$, so by Lemma 3.2.3, we have a path decomposition $f'(\Sigma_2)$ of G_2 of cardinality $|f'(\Sigma_2)| = |\Sigma_2| \leq \frac{1}{2}\mu + \lfloor \frac{2}{3} g \rfloor$. And there is at least one path in $f'(\Sigma_2)$ that has y as an endpoint.

Remove the edge wz from $f'(\Sigma_2)$, we have a path decomposition $f'(\Sigma_1)$ of G_1 of cardinality $|f'(\Sigma_1)| \leq |f'(\Sigma_2)| \leq \frac{1}{2}\mu + \lfloor \frac{2}{3} g \rfloor$. And there is at least one path in $f'(\Sigma_1)$ that has y as an endpoint.

We now reintroduce the edge xy without disturbing the cardinality of the path decomposition $f'(\Sigma_1)$ of G_1. Since each vertex adjacent to x in G_1 is odd and since y is the endpoint of at least one path of $f'(\Sigma_1)$. After completing Lovász construction about x with respect to y, we obtain a path decomposition $f'(\Sigma)$ of G such that $\wp(G) \leq |f'(\Sigma)| = |f'(\Sigma_1)| \leq \frac{1}{2}\mu + \lfloor \frac{2}{3} g \rfloor$.

This completes the proof of the main theorem.

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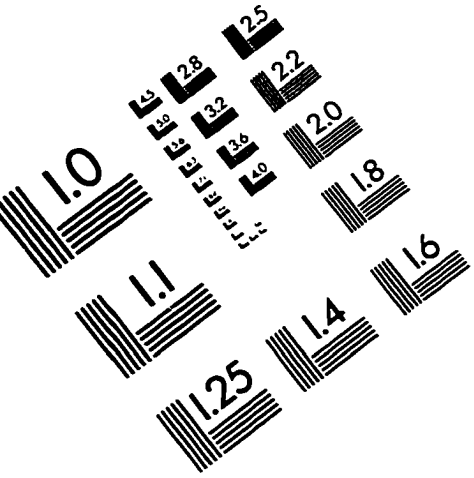
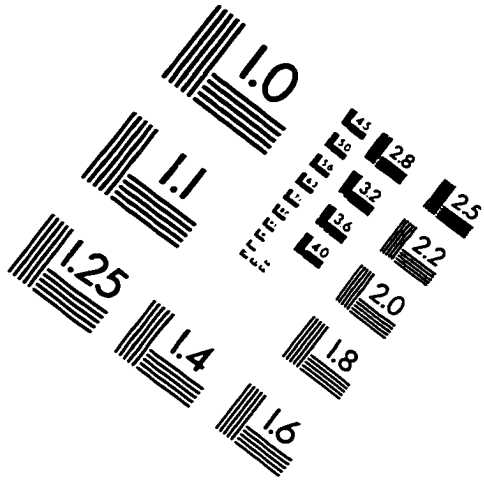
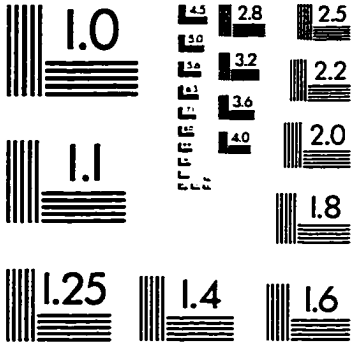
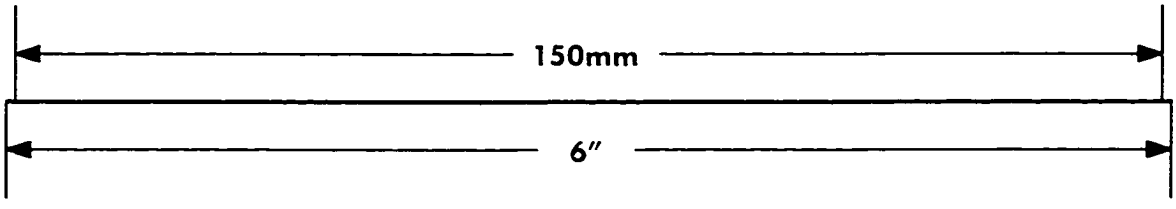
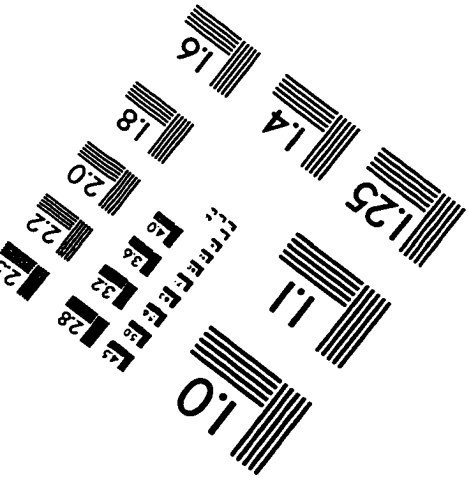
IMAGE EVALUATION TEST TARGET (QA-3)











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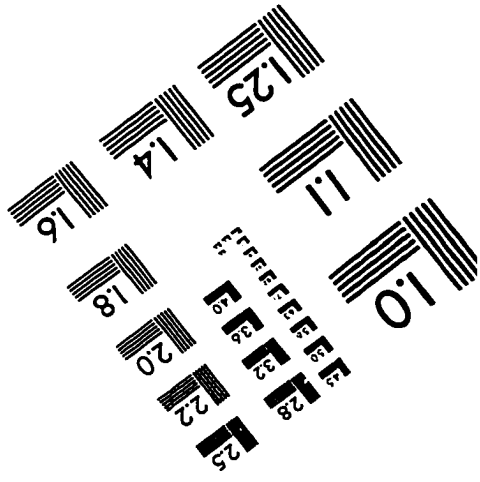

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